Vidyalankar Institute of Technology

Department of Computer Engineering

DEVOPS MINI PROJECT

Week 2 Report

Automated Grocery Inventory Dashboard

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1. Agile Methodology

The Automated Grocery Inventory Dashboard will be developed using the Agile methodology. Instead of building the entire system at once, the project is divided into small weekly iterations based on the 15-week project schedule. Each week focuses on completing a specific milestone, allowing continuous development, testing, and improvement.

Using Agile makes the development process more flexible and organized. Features are implemented gradually, feedback can be incorporated quickly, and the project remains aligned with the planned timeline. This approach also supports DevOps practices by enabling continuous integration, testing, and deployment throughout the project lifecycle.

1.1 Benefits of Agile

- Incremental development of features.
- Better progress tracking.
- Easier testing and debugging.
- Continuous integration with DevOps tools.
- Reduced project risk.

2. User Stories

User stories describe the system from the user's perspective and help identify the core functionalities of the Grocery Inventory Dashboard.

US-01: Add Grocery Item

As a store staff member,
I want to add a new grocery item,
So that it becomes available in the inventory.

US-02: Update Grocery Item

As a store staff member,
I want to update product details,
So that inventory information remains accurate.

US-03: Delete Grocery Item

As a store staff member,
I want to remove discontinued products,
So that the inventory remains organized.

US-04: Record Stock In

As a store staff member,
I want to add newly received stock,
So that inventory quantities are updated.

US-05: Record Stock Out

As a store staff member,
I want to reduce stock when products are sold,
So that inventory always reflects the available quantity.

US-06: Search Products

As a user,
I want to search products by name or category,
So that I can quickly find required items.

US-07: View Dashboard

As a store owner,
I want to view inventory statistics,
So that I can monitor stock availability.

US-08: Receive Stock Alerts

As a store owner,
I want to receive low-stock and out-of-stock alerts,
So that products can be restocked on time.

3. Acceptance Criteria

The acceptance criteria define the conditions that must be satisfied for each user story to be considered complete.

Add Grocery Item

- User enters all required product details.
- Item is added successfully to the inventory.
- New item appears in the inventory list.

Update Grocery Item

- Existing product information can be modified.
- Updated details are saved successfully.
- Changes are reflected immediately.

Delete Grocery Item

- User can remove an existing product.
- Deleted item no longer appears in the inventory.

Stock Transactions

- Stock In increases available quantity.
- Stock Out decreases available quantity.
- Updated stock is displayed on the dashboard.

Search Products

- Users can search products by name.
- Users can filter products by category.
- Matching products are displayed correctly.

Inventory Alerts

- Low-stock products are highlighted.
- Products with zero quantity are marked as Out of Stock.

4. Product Backlog

The product backlog contains all planned features for the Grocery Inventory Dashboard arranged according to their priority.

Priority	User Story	Status
High	Add Grocery Item	Pending
High	Update Grocery Item	Pending
High	Delete Grocery Item	Pending
High	Stock In Transaction	Pending
High	Stock Out Transaction	Pending
High	Dashboard	Pending
Medium	Search Products	Pending
Medium	Inventory Alerts	Pending
Medium	Jenkins CI Pipeline	Pending
Medium	Selenium Testing	Pending
Medium	Docker Deployment	Pending
Medium	Ansible Automation	Pending

5. Sprint Plan

The project is divided into weekly sprints based on the faculty timeline.

Week	Sprint Goal
1	Problem Definition and Project Scope
2	Agile Planning and DevOps Workflow
3	Requirements Analysis and Architecture Design
4	GitHub Repository Initialization
5	Feature Development using Git Branching
6	MVP Completion and Git Collaboration
7	Jenkins Continuous Integration
8	Pipeline as Code and Deployment
9	Selenium Test Development
10	Continuous Testing in Jenkins
11	Docker Containerization
12	Jenkins-Docker Continuous Deployment
13	Configuration Management using Ansible
14	Automated Provisioning and Validation
15	Final Demonstration and Documentation

6. Task Board (Kanban)

The Kanban board represents the current status of project activities.

To Do	In Progress	Done
Add Grocery Item	Agile Planning	Problem Definition
Update Grocery Item	Product Backlog	Stakeholders
Stock Transactions	User Stories	Project Objectives
Dashboard	Acceptance Criteria	MVP Scope
Search	Sprint Planning	Week 1 Report
Jenkins Setup		
Docker Deployment		
Ansible Automation		

7. Definition of Done (DoD)

A task is considered complete only when it satisfies the following conditions:

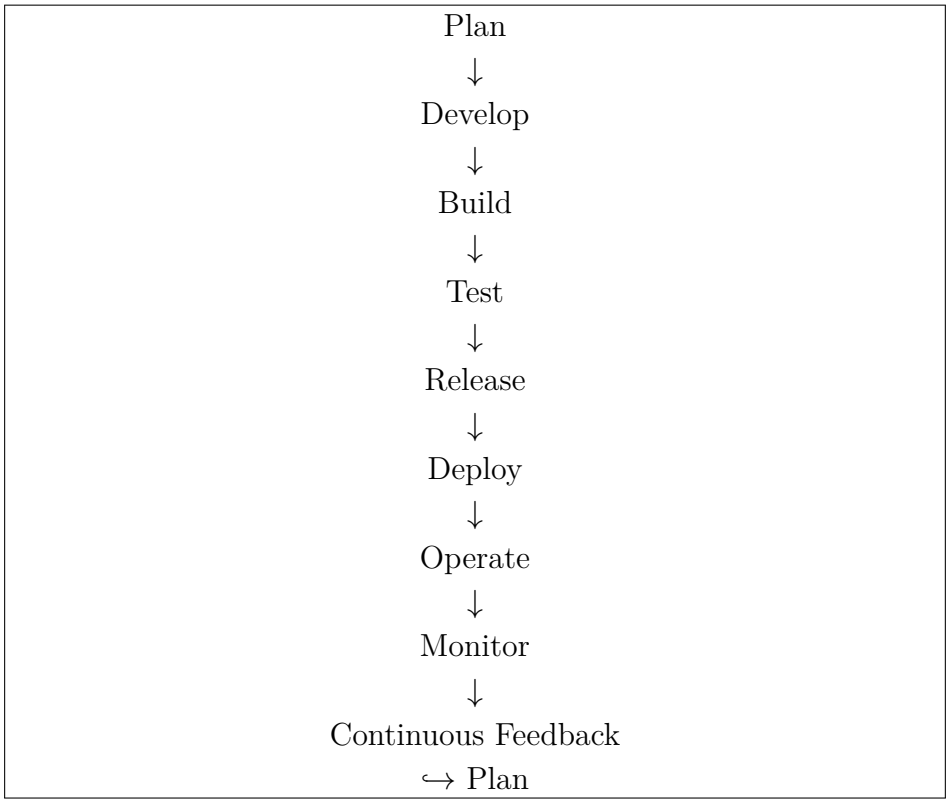
- The assigned functionality is implemented successfully.
- The code compiles and runs without errors.
- Basic testing is completed.
- The changes are committed to Git using meaningful commit messages.
- The acceptance criteria for the user story are satisfied.
- Documentation is updated whenever required.

Following these conditions ensures that every completed feature is stable, testable, and ready for integration into the project.

8. DevOps Lifecycle

The Grocery Inventory Dashboard will be developed following the DevOps lifecycle to ensure continuous development, testing, deployment, and monitoring.

DevOps Workflow



Plan: Gather requirements, prepare backlog, and define sprint goals.
Develop: Implement application features using Git and feature branches.
Build: Compile the project automatically using Maven and Jenkins.
Test: Execute Selenium test cases to validate application functionality.

Release: Prepare a stable version for deployment.
Deploy: Deploy the application using Docker containers.
Operate: Run and maintain the deployed application.
Monitor: Monitor application health and use feedback for future improvements.